

SPPU IoT Unit 1 Notes

1. IoT Definition and Characteristics

Introduction: The Internet of Things (IoT) connects physical objects such as sensors, machines, vehicles, home appliances, and wearable devices to the Internet, enabling them to collect, exchange, and process data without human intervention.

Definition: IoT is a network of physical devices embedded with sensors, software, and communication technologies that allow them to connect and exchange data over the Internet.

Characteristics: Connectivity, Intelligence, Dynamic Nature, Heterogeneity, Scalability, Security, and Self-Configuration.

2. Functional Blocks of IoT Ecosystem

Functional blocks include Devices, Communication, Services, Management, Security, and Applications. These blocks work together to provide intelligent IoT services.

3. IoT Communication Models

Device-to-Device (D2D), Device-to-Cloud, Device-to-Gateway, and Back-End Data Sharing.

4. IoT Protocols

MQTT: Lightweight publish-subscribe protocol.

CoAP: Lightweight REST-based protocol using UDP.

HTTP: Standard request-response web protocol.

5. M2M Communication

Machine-to-Machine communication refers to automatic exchange of information between machines without human intervention.

6. Evolution of IoT

RFID → Wireless Sensor Networks → Internet Connectivity → Cloud Computing → Big Data Analytics → Artificial Intelligence → Modern IoT.

7. IoT Challenges

Security, Privacy, Interoperability, Scalability, Data Management, Energy Consumption, Reliability, and Legal Issues.

8. Wireless Sensor Network (WSN)

A network of autonomous sensor nodes that sense environmental conditions and communicate wirelessly to transfer data to a central location.

9. Cloud Computing in IoT

Provides storage, processing, analytics, and remote access for large volumes of IoT data.

10. Big Data Analytics in IoT

Analyzes massive IoT-generated datasets to discover patterns, trends, and useful insights.

11. Embedded Systems in IoT

Combination of hardware and software designed to perform a specific function within a larger system.

12. IoT Enabled Applications

Smart Homes, Smart Healthcare, Smart Agriculture, Smart Cities, Industrial IoT, and Smart Transportation.

Note: This PDF contains the topics requested from the conversation and serves as a Unit-1 revision document.